Arcadia

Multimodal Adaptive Mastery Engine

RAG-Powered · AdaptiveQuizzing · Multilingual Chatbot · Cheatsheets · Flashcards · Jargon Breakdown

Institue: Manipal Institute of Technology	**Team:** Echelon (Solo Member)
Challenge: Track 3 — AI Study Buddy	**Email:** nityant.mitmpl2024@learner.manipal.edu

THE TEAM

Team Echelon is a solo developer at Manipal Institute of Technology building technology that democratises education. Inspired by students in vernacular-medium schools who struggle to bridge classroom notes and exam performance, this project proposes a focused, credible AI platform: a **Multimodal Adaptive Mastery Engine** that combines RAG-based knowledge retrieval, adaptive quizzing, and multilingual accessibility covering every subject from Coding and Maths to Biology, Arts, and Philosophy for every student, in the language they think in.

THE CONCEPT

Arcadia is a full-stack multimodal AI study platform that transforms handwritten notes and uploaded documents into a personalised, multilingual learning experience. The core engine is built around three pillars:

(1) RAG-Based Knowledge Retrieval: any uploaded note or PDF becomes a queryable knowledge base.

(**2) Adaptive Mastery Quizzing:** 3-tier quizzing that adapts to the student's demonstrated understanding.

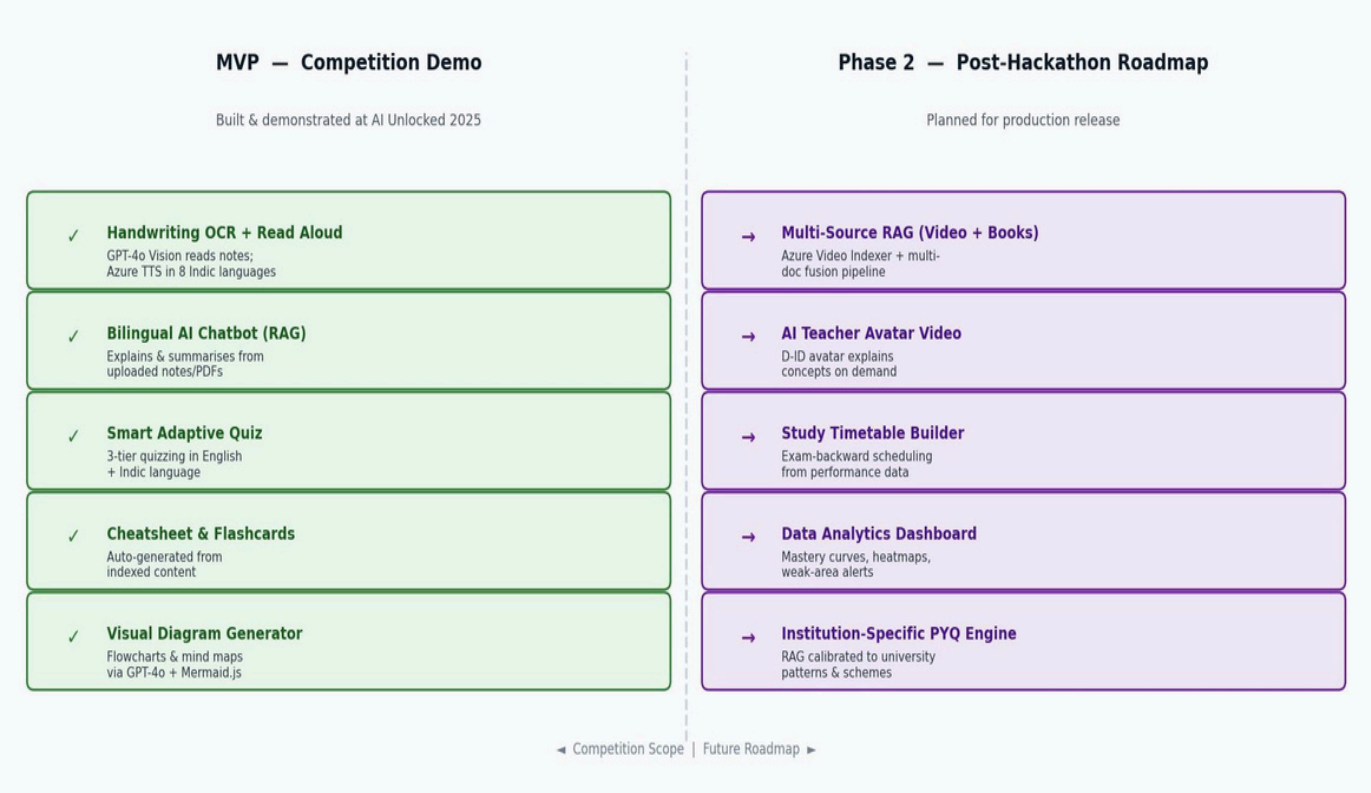
(3) Multilingual Accessibility: all content read aloud and explained in 8 Indic languages via Azure AI Speech and Translator.

The platform solves five real problems: notes never revised; complex concepts stay opaque; vernacular -medium students disadvantaged by English -first EdTech; resources scattered; exam prep lacking institution -specific context.

CORE FEATURES (COMPETITION MVP + PHASE 2 ROADMAP)

To maintain a credible and demonstrable scope for AI Unlocked 2025, Arcadia separates what will be **built and demonstrated at the competition** from what is planned for post-hackathon development.

Fig 1: MVP scope (competition demo) vs. Phase 2 roadmap — clear separation of delivered vs. planned features



TARGET AUDIENCE & MARKET SIZE

Primary: 250M+ secondary students Classes 8–12 (UDISE+ 2023).

Secondary: 58M undergraduates (AISHE 2022) + 85M competitive exam aspirants (FICCI 2023).

Tertiary: 60M self-learners, tutors, and upskilling professionals across all domains.

Platform is mobile-first Android (92% India market share) with PWA for desktop. 8 Indic languages at launch.

Total Addressable Market: 500M+ learners in India.

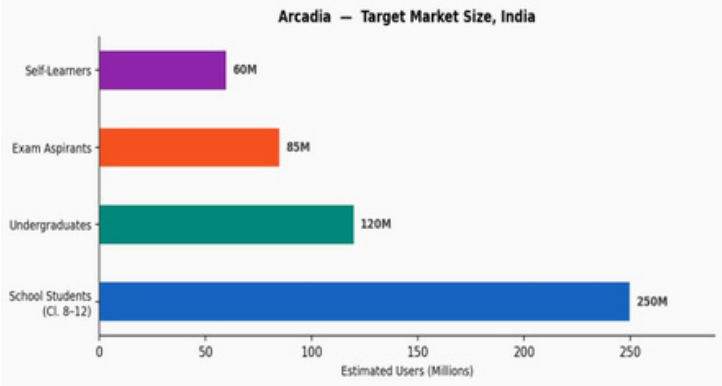


Fig 2: Target market segments India (UDISE+2023,AISHE2022,FICCIEdTech2023)

USER PERSONAS

Riya,16, Class11 Science, CBSE, Bhopal.

Writes detailed but messy Hindi notes during Physics lectures. Struggles with English textbooks. Needs bilingual TTS read-aloud, concept cheatsheets, flashcard drills, and adaptive quizzing in Hindi.

Core pain: language gap + no engaging revision workflow.

Arjun, 21, B.Tech CSE Sem 4, MIT Manipal.

Photographs whiteboards but never revises. Uploads PDFs and asks the chatbot to explain concepts he doesn't understand. Uses the adaptive quiz to track weak areas before exams.

Core pain: scattered resources, no personalised mastery tracking.

MEASURABLE OUTCOMES — HOW WE DEFINE SUCCESS

Arcadia's impact is measurable, not aspirational. The following target metrics are based on published research on spaced repetition (Ebbinghaus, 2014), adaptive learning systems (Pane et al., RAND 2015), and multilingual comprehension studies (NCERT 2022). These will be validated through a **structured pilot study at MIT Manipal** with 50–100 students across 4 weeks, using pre/post assessment and session engagement tracking.

40%	50%	3×	85%	60%
Improvement in quiz retention	Reduction in revision time per topic	More study sessions/weekvs passive reading	Students report better concept clarity	Reductionin jargon confusion(pre/post test)

Fig 3: Target impact metrics for the Arcadia pilot study grounded in educational research benchmarks

HOW IT WORKS — 7-STEP MVP PIPELINE

Upload	Student uploads handwritten notes photo or PDF. **GPT-4o Vision** reads and extracts handwritten text. **Azure Form Recogniser** parses structured PDFs into clean text chunks.
RAG Index	Extracted text is embedded and indexed in **Azure AI Search (vector store)** via LangChain. Each chunk is tagged with subject and topic. The index is user-specific and persists across sessions.
Localise	**Azure AI Translator** converts all content into the student's chosen Indic language. **Azure AI Speech (neural TTS)** reads concept cards aloud. Bilingual toggle shows English + regional side-by-side.
Generate	GPT-4o generates the core learning suite from indexed content: **concept cards**, **one-page cheatsheet**s, **spaced-repetition flashcard decks**, and **visual diagrams/flowcharts** via Mermaid.js rendering.
Quiz	**3-tier adaptive quiz engine:** Tier 1 = recall (definitions, facts), Tier 2 = application (problem-solving), Tier 3 = analysis (evaluation, synthesis). Questions generated in English and/or Indic language. Wrong answers trigger targeted re-explanation of that specific concept before moving on.
Chat	**Multilingual RAG chatbot** (GPT-4o + Azure AI Search): student asks "explain Newton's 2nd law in Hindi" or "summarise Chapter 4" and gets a contextually accurate answer pulled from their own uploaded notes. Azure Redis maintains session context.
Track	Session performance — quiz scores, flashcard accuracy, topic mastery — is stored in **Azure Cosmos DB**. Arcadia computes a lightweight mastery confidence score per concept using a weighted combination of quiz accuracy, response latency, attempt count, and spaced-repetition interval decay. This score dynamically determines revision priority, flashcard resurfacing frequency, and quiz difficulty escalation.

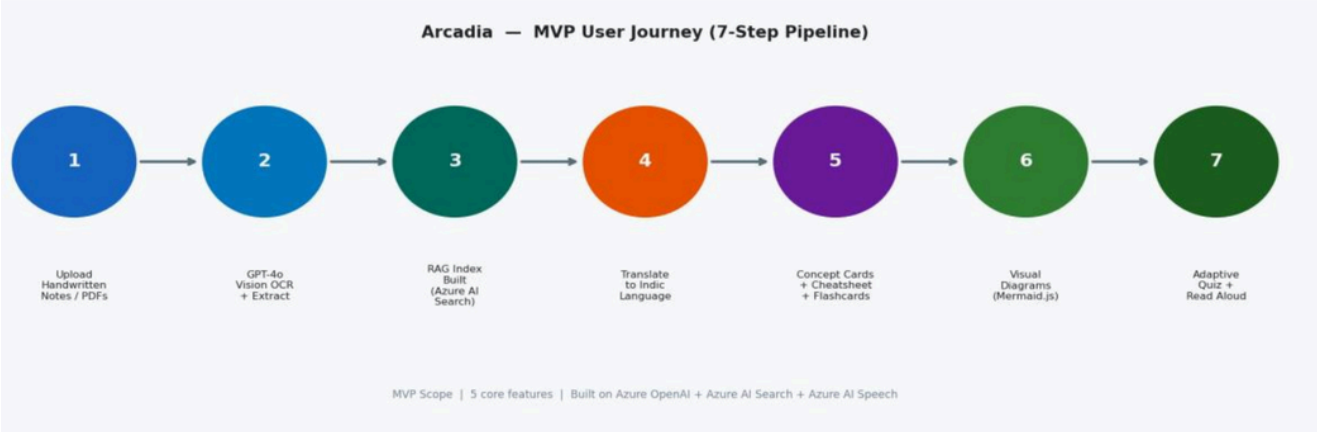


Fig 4: Arcadia MVP — 7-step user journey from upload to adaptive mastery tracking

SYSTEM ARCHITECTURE (MVP + PHASE 2)

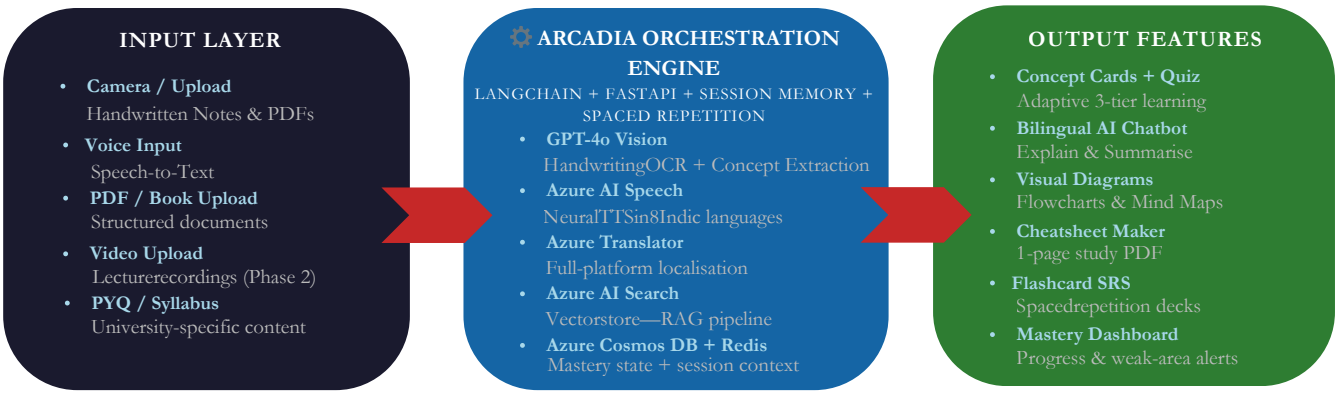


Fig 5: Arcadia system architecture — Azure AI services orchestrated by LangChain + FastAPI backend

CORE TECHNOLOGIES (MVP STACK)

Technology	Role in Arcadia MVP	Why It Is Critical
Azure OpenAI — GPT-4o Vision	Handwriting OCR, concept extraction, cheatsheet + flashcard generation, bilingual chatbot, visual diagram creation	Multilingual + vision + reasoning in one API. The central intelligence layer of Arcadia's MVP.
Azure AI Search (Vector Store)	RAG pipeline: embed, index, and retrieve from all uploaded notes and PDFs. Enables context-aware chatbot answers.	Makes every uploaded document instantly queryable. Core to the "your own notes" RAG experience.
Azure AI Speech — Neural TTS	Reads concept cards, cheatsheets, and quiz questions aloud in English and 8 Indic languages using neural voices.	Hindi, Tamil, Telugu, Marathi voices the accessibility cornerstone. Non-negotiable for vernacular learners.
Azure AI Translator	Full-platform localisation: all extracted content, concept cards, quiz questions translated to chosen Indic language.	Domain-accurate STEM and humanities vocabulary. Enables true bilingual side-by-side learning mode.
LangChain + FastAPI (Python)	RAG orchestration, spaced-repetition logic, session memory management, multi-step GPT-4o reasoning chains.	Industry-standard LLM application framework. Clean, modular, testable — easy to extend for Phase 2 features.
Azure Cosmos DB + Redis Cache	Student profile, quiz scores, flashcard mastery state, topic confidence scores; chat session context window.	Globally distributed low-latency database. Redis enables sub-millisecond conversation state retrieval.
React Native (Expo) + Mermaid.js	Cross-platform mobile app (Android + iOS). Inline visual diagram rendering for flowcharts and mind maps.	Single codebase targets Android (92% India market). Mermaid renders GPT-4o generated diagrams live in app.
Azure App Service + Blob + CDN	Backend API hosting; audio file storage for TTS output; global CDN delivery of media to mobile clients.	Auto-scales for exam-season traffic spikes. Cost-efficient pay-per-use model suitable for student-priced tiers.

BUSINESS PLAN — PILOT-FIRST, SUSTAINABLY SCALABLE

Phase	Timeline	Focus & Target
Pilot	Month 1–3	50–100 students at MIT Manipal. Core MVP: OCR + RAG chatbot + quiz + cheatsheets. Measure pre/post retention improvement.
MVP Launch	Month 4–6	Full MVP on Android. Freemium live. 5 institutional pilots. Target: 1,000 MAU, Rs. 20,000 MRR, validated impact metrics.
Growth	Month 7–12	B2B school licenses (Rs. 999/yr per class). Institution-specific RAG. Target: 20 schools, 10,000 MAU, Rs. 1.5L MRR.
Phase 2	Month 13–18	AI teacher video, video-source RAG, analytics dashboard, timetable builder. Target: Scalable beyond pilot with validated unit economics.

Freemium SaaS (Individual): Free tier — 20 scans/month, English only, basic quiz. **Arcadia Plus** (Rs. 79/month or Rs. 649/year): unlimited scans, all 8 Indic languages, TTS read-aloud, cheatsheets, flashcard decks, adaptive quizzing. **Arcadia Pro** (Rs. 149/month): everything + full RAG chatbot + mastery tracking dashboard.

B2B Institutional License (Rs. 999/year per classroom, ~30 students): teacher dashboard, class-wide mastery analytics, institution-specific RAG loaded with course syllabus and past papers. Pilot target: MIT Manipal + 4 partner colleges in Phase 1.

Phase 2 Revenue Expansion: AI teacher video add-on (Rs. 29/video or bundled). White-label API for EdTech platforms. Government EdTech tender opportunities (NEP 2020 multilingual education mandate creates regulatory tailwind).

TRACK ALIGNMENT + ADDITIONAL INFORMATION

Track3Alignment:

Arcadia directly alignswith and extends theTrack 3 vision.

Sees messy handwritten notes — GPT-4o Vision reads and understands handwriting from a photo.

Reads them aloud — Azure AI Speech with neural Indic voices.

Breaks down complex concepts — RAG chatbot explains, simplifies, and generates visual diagrams from the student's own notes.

Quizzes in English and a favourite Indian language — 3-tier adaptive quiz engine with full Azure Translator localisation, making studying engaging rather than exhausting.

Additional MVP features (cheatsheets, flashcards, bilingual chatbot, visual diagrams) go beyond the brief while remaining fully buildable by a solo developer within the hackathon window. AI teacher video and multi-source RAG are Phase 2, keeping the competition

MVP credible and focused. Arcadia closes the cognitive loop from perception (vision input) → comprehension (RAG explanation) → reinforcement (adaptive quizzing) → retention (spaced repetition), creating a complete AI-driven mastery cycle rather than a passive chatbot experience.

MVP Tech Stack:

React Native (Expo) · FastAPI · LangChain · Azure OpenAI GPT-4o · Azure AI Search · Azure AI Speech (Neural TTS) · Azure AI Translator · Azure Form Recogniser · Azure Cosmos DB · Azure Redis Cache · Azure Blob Storage · Azure App Service · Mermaid.js · GitHub Actions CI/CD

References:

UDISE+ Annual Report 2022–23 | AISHE Report 2021–22 | FICCI-EY "Future of Indian EdTech" 2023 | Pane et al., RAND Corp. 2015 (Adaptive Learning) | NEP 2020 Multilingual Education Policy | learn.microsoft.com/en-us/azure/ai-services